Algebraic World Model: From Exploration to Proofs

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Abstract

AI has made remarkable progress in mathematics, solving challenging problems with machine-checkable proofs in Lean, yet typically relying on predefined mathematical definitions and formal specifications. This raises a natural question: can it reason rigorously about an unknown world? We introduce the Algebraic World Model (AWM), a framework that turns interaction with unfamiliar environments into explicit mathematical models supporting formal reasoning. AWM explores the environment and develops executable Python hypotheses, continually testing and revising them against observations. It then structures these hypotheses into typed objects and shared rules, simplifies their structure through category-theory-inspired transformations, and formalizes the resulting model in Lean 4 to prove reachability and impossibility.

AWM can formally prove that a goal is unreachable, even beyond explored trajectories. On two impossible game levels, it constructs closed, kernel-checked Lean proofs excluding every legal winning sequence in all six runs, while Claude Opus 5.5 and GPT-6 Astra produce none in twelve baseline runs. Four of AWM's six initial bounds match the minimum winning costs in the games, compared with one for Opus 5.5 and none for GPT-6 Astra; feedback-guided refinement brings AWM to six of six. Moreover, AWM makes learned rules explicit and checkable. Across all 25 public ARC-AGI-3 games, Lean replays the winning trajectories of all 183 levels against AWM's models. Beyond replay, frozen rules predict all 4,160 tested transitions on unseen layouts and reject all 4,031 after a mechanic change. Built on Claude Opus 5, AWM raises its score from 40.7 to 99.9; its certified solutions use 43% of the human baseline's actions.

1 Introduction

Artificial intelligence has made remarkable progress in mathematical reasoning: automated systems now produce Lean proofs of olympiad problems and of open Erdős problems (Achim et al., 2025; Hubert et al., 2026; Sothanaphan, 2026; Tsoukalas et al., 2026). These advances invite a broader question:

Can an agent reason rigorously about an unknown world?

In formal mathematics the definitions needed to state a problem are given in advance, and a proof checker verifies whether a conclusion follows from them. In an unfamiliar environment the agent must first discover the relevant objects, infer how actions transform them and determine what counts as success; the rules required for formal reasoning are themselves hypotheses about the world, learned from experience and open to error.

Euler's treatment of the Königsberg bridges shows what such a structure can provide (Euler, 1741). Land masses become vertices and bridges become edges. A walk crossing every bridge exactly once

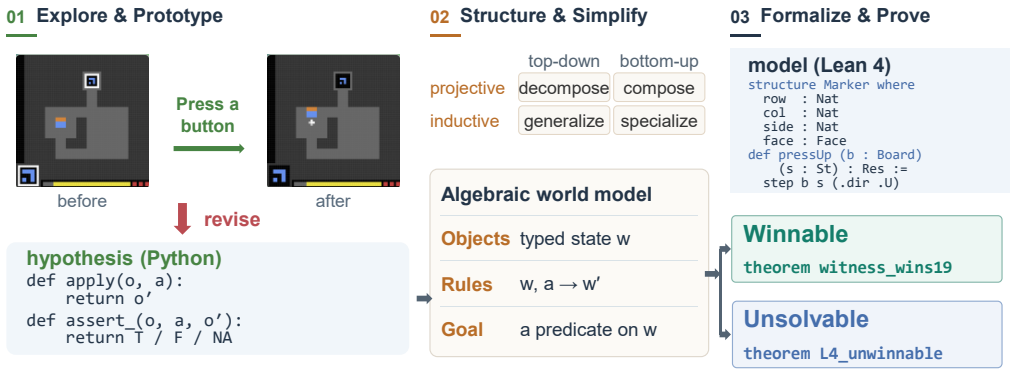


Figure 1: AWM's three-stage pipeline: interactive exploration builds a provisional executable model, algebraic transformations reveal and simplify its structure, and Lean 4 formalization enables certified proofs of reachability or impossibility.

pairs arrivals with departures at each intermediate vertex, so the graph can have at most two odd-degree vertices; the Königsberg graph has four. The graph rules out every candidate walk, including the walks nobody tried. Finding one winning route shows that a goal can be reached; showing that none exists needs a model of this kind, and in an unfamiliar environment that model has to be learned.

The Algebraic World Model (AWM) builds such a model through three stages (Figure 1). **Explore & Prototype:** the agent actively explores the environment, building and iteratively revising a provisional executable model from observed transitions. **Structure & Simplify:** it organizes this prototype into typed objects and rules, then applies category-theory-inspired transformations to discover and simplify their underlying structure while preserving empirically tested distinctions. **Formalize & Prove:** it encodes the model in Lean 4 (de Moura & Ullrich, 2021), and constructs proofs of reachability, impossibility, and other stated properties, checked by Lean's small trusted kernel. A winning witness certifies reachability; an invariant or a suitable cost lower bound can exclude every legal winning sequence under the model.

In other words, AWM can formally prove that a task is impossible under its learned model. We evaluate this ability on ARC-AGI-3 (ARC Prize Foundation, 2026b), whose games reveal neither their rules nor their goals. We construct two impossible variants of `sc25` levels by preserving their mechanics and setting action budgets just below the minimum winning costs (§4). With three runs per level for each agent, AWM constructs closed, kernel-checked Lean proofs excluding all legal winning sequences under its learned models in all six runs, while Claude Opus 5.5 and GPT-6 Astra produce no such proofs in their twelve baseline runs.

The difference extends beyond proof construction. AWM's initial bounds match the minimum winning costs in four of six runs, compared with one for Opus 5.5 (no proof) and none for Astra; the remaining baseline runs report incorrect bounds or remain inconclusive. Feedback that a cheaper win exists brings AWM to six of six correct bounds, with new closed proofs under the refined models. For example, on level 4 Astra claims impossibility with a minimum winning cost of 22 without proving optimality, whereas an independently rebuilt AWM model proves the correct bound of 19.

Moreover, AWM makes learned mechanisms explicit and interpretable through typed objects and named rules, allowing their behavior to be checked step by step. Across all 25 public ARC-AGI-3 games, AWM builds compact models with a median of six object types and ten rules. Lean replays all 183 winning trajectories, checking the modeled state transitions and the final goal conditions (§3). With Opus 5, AWM clears all 25 games at a score of 99.9, competitive with leading code-executing systems; its certified solutions use 43% of the human baseline's actions.

We make three contributions.

- **A framework from exploration to algebraic structure.** AWM turns interaction with an unknown environment into executable hypotheses, then discovers shared mathematical structure through category-theory-inspired transformations. The resulting compact model expresses related

mechanisms as reusable, compositional rules, enabling unified reasoning about their behavior (§2).

- **Formal impossibility reasoning in unknown worlds.** AWM constructs closed, kernel-checked Lean proofs that rule out every legal winning sequence under its learned model and specified constraints, including sequences never explored. This moves beyond empirical judgments based on finite play to formally certified conclusions under explicit learned premises (§4).
- **Inspectable models and traceable proofs.** AWM represents environmental dynamics through explicit typed objects and named rules, whose predictions can be checked against observed transitions. Its formal representation supports step-by-step replay and exposes the premises behind each proof, making learned mechanisms interpretable and discrepancies traceable to their underlying assumptions (§3; §4.3).

2 Method

AWM aims to turn interaction with an unknown environment into a mathematical model that supports rigorous reasoning about both reachability and impossibility. This requires more than collecting rules that explain observed transitions: the rules must form a coherent representation of the environment, and that representation must support reasoning about action sequences the agent has never explored.

We first describe the target model (§2.1), then present the three stages that construct and use it (Figure 1): Explore & Prototype develops executable hypotheses through interaction; Structure & Simplify organizes them into a compact algebraic model; and Formalize & Prove completes the model in Lean 4 and establishes machine-checked results.

2.1 The Algebraic World Model

An algebraic world model describes an environment through typed objects, rules, and a goal predicate, represented in executable Python and formal Lean 4 to support both empirical testing and mathematical reasoning.

Objects. Object types define the structure and validity of states, providing a common vocabulary for describing the environment. A parser maps observations, together with any required history or auxiliary information, to typed states w.

Rules. Rules describe how actions transform these states. Each Python rule specifies an applicability scope and a checker that reports agreement, contradiction, or non-applicability on observed transitions; effect rules also predict successor states or their components. Together, the rules define legal actions, their effects and costs $\mathrm{cost}(w, a)$, while the goal predicate identifies successful states.

Related mechanisms can share a parameterized operation $r(\theta)$, with θ determined by the state. This allows common properties to be expressed and proved uniformly across different mechanisms.

Formal representation. The same model is formalized in Lean 4, with objects and rules encoded as definitions and properties of their dynamics expressed as theorems. Its executable representation supports testing against observations, while its formal representation supports kernel-checked proofs over legal action sequences (§2.4).

2.2 Explore & Prototype

Setting. We consider deterministic, goal-directed environments with partial observations, modeled in the POMDP setting (Kaelbling et al., 1998). At step t, the agent observes o_t, takes an action a_t, and receives o_{t+1} without access to the underlying state or transition function. Its experience forms a corpus $\mathcal{D}_t$ of distinct transitions $x = (o, a, o')$.

Building and testing hypotheses. AWM actively explores the environment and maintains a provisional executable model as a *concept book*. Each concept is a Python hypothesis about an environmental mechanism, containing a natural-language claim, an applicability scope, supporting transitions, a checker, and an optional predictor.

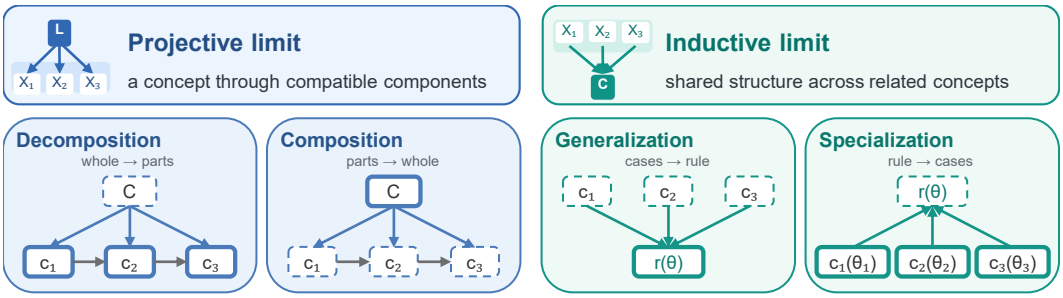


Figure 2: Four transformations inspired by projective and inductive limits, applied to object types and rules. Dashed boxes mark removed components and bold boxes mark introduced components; arrows indicate relationships between components. Generalization uses $r(\theta)$, and an instance is written $r(\theta_i)$; specialization makes distinctions explicit through case-specific types or parameters.

For a concept c, its checker returns $v_c(x)$: `TRUE` if the transition agrees with the hypothesis, `FALSE` if it contradicts it, and `NA` if it falls outside its scope. The optional predictor specifies expected action effects for comparison with subsequent observations.

After each action batch, AWM rechecks the concept book $\mathcal{C}_t$ against the expanded corpus $\mathcal{D}_t$. Further interaction is allowed only when

$$\forall c \in \mathcal{C}_t,\ \forall x \in \mathcal{D}_t : \quad v_c(x) \notin \{\mathsf{FALSE}, \mathsf{ERR}\}, \tag{1}$$

where ERR denotes checker failure, including exceptions and timeouts.

This recheck ensures that contradicted or failing concepts must be revised or withdrawn before exploration continues. The agent then keeps revises and uses its hypotheses to plan toward inferred goals or probe competing explanations.

The resulting prototype captures tested mechanisms but may describe related behaviors independently, motivating the next stage.

2.3 Structure & Simplify

The concept book may contain fragmented, overlapping descriptions of environmental mechanisms. Structure & Simplify turns these hypotheses into a compact model by identifying shared structure and reorganizing components through their relationships.

AWM first extracts the required object types, constructs a parser from observations to typed states, and rewrites concepts as rules over these objects. It then restructures these *components*—object types and rules—through four category-theory-inspired transformations. A *projective limit* assembles compatible descriptions into a coherent whole, while an *inductive limit* glues objects along shared structure (Mac Lane, 1998; Kashiwara & Schapira, 2006; Yuan, 2023). These two perspectives motivate transformations based on compatibility and shared structure, respectively (Figure 2).

Following projective limits, *decomposition* expresses a complex component through simpler components and their compatibility relations. Conversely, *composition* expresses a component through compatible existing components, moving the composite into the derived layer when a separate primitive definition is unnecessary.

Following inductive limits, *generalization* extracts shared structure from multiple components, expressing source rules c_i through a parameterized operation $r(\theta_i)$ with state-dependent θ_i. Conversely, *specialization* refines overly general components into more specific cases or introduces parameters to reduce exceptions and redundancy.

Preserving tested distinctions. Every transformation must preserve verdicts on recorded transitions. For rules, AWM also uses a diagnostic set $\mathcal{N}$ of perturbed recorded transitions to check whether simplification loses previously detected rejections.

Define $\mathrm{Rej}(c) = \{x \in \mathcal{N} : v_c(x) = \mathsf{FALSE}\}$ and $\mathrm{Rej}(S) = \bigcup_{c \in S} \mathrm{Rej}(c)$. Replacing rules S with T must preserve coverage of recorded transitions and satisfy

$$\mathrm{Rej}(T) \supseteq \mathrm{Rej}(S). \tag{2}$$

For object types, which have no checkers, transformations are evaluated by distinctions between recorded states, using state pairs as witnesses, and by rule verdicts once rules are available.

AWM applies these transformations and tests to object types, then rules, and finally rechecks object types against the settled rules. The resulting compact structure provides a common representation for formal reasoning.

2.4 Formalize & Prove

AWM completes the model in Lean 4, encoding object types as structures or inductive types, rules as `def` definitions, and derived properties as `theorem` statements. The transition function dispatches to named rules, while unresolved assumptions remain explicit theorem hypotheses rather than unchecked `axiom` declarations. Replaying recorded winning trajectories, the Lean model reproduces observed behavior on every component it represents, grounding its formal definitions in the dynamics the agent encountered. This enables theorems about all legal action sequences under the learned model, extending formal reasoning beyond the trajectories actually explored.

Proving reachability and impossibility. A level is *winnable* in the model if a legal action sequence reaches the goal. AWM proves reachability by constructing a witness checked in Lean. Impossibility can arise because the goal is unreachable, or because reaching it requires more cost than the available budget. AWM proves the former using an invariant that excludes the goal, and the latter using a lower bound on the cost of any winning sequence. For budgeted tasks, a potential V estimates the minimum remaining cost from each state and satisfies

$$V(w) \le \mathrm{cost}(w, a) + V(w') \quad \text{for every legal transition,} \qquad V(w) \le 0 \quad \text{on the goal,} \tag{3}$$

so every winning sequence costs at least $V(w_0)$, proving impossibility when this exceeds the budget. All proofs are closed and checked by the Lean kernel, without unproved placeholders (`sorry`) or kernel-bypassing computation (`native_decide`), establishing rigorous guarantees under the learned model's explicit premises (definitions and the transfer condition in Appendix A).

2.5 Example: one rule for four mechanisms

In game `lp85`, buttons move colored tiles, and a level is won when every slot holds its matching color. During Explore & Prototype, the agent discovers eleven concepts describing four button mechanisms: belts, stars, hubs, and panels (Figure 3a). In Structure & Simplify, AWM applies the generalization transformation to recognize that all four mechanisms permute tile positions, replacing the eleven concepts with a shared operation, `press_permutes`, parameterized by the pressed button. On the diagnostic set used for this merge, the shared rule retains all 55 rejections of the source concepts and rejects 21 additional corrupted transitions. The shared rule also shortens reasoning: because one color-count invariant covers every button, proofs that a variant is unsolvable need two to three times fewer lemmas than proofs over the eleven separate concepts (Appendix B.3). A separate check tests whether such a model-level proof also holds for the game: on the level-6 tile-swap variant U, Lean verifies its unsolvability proof against an engine-derived graph covering all 52,500 transitions among 7,500 configurations, with no axioms beyond Lean's standard ones (Appendix B.4).

In Formalize & Prove, Lean certifies a five-click winning sequence (Figure 3b) and proves a recolored variant unwinnable using the color-count invariant shared by all button permutations (Figure 3c). The final model contains six object types and six rules, and Lean replays all 199 recorded transitions without mismatch.

3 Twenty-five public games

ARC-AGI-3 (ARC Prize Foundation, 2026b) scores each level by the squared ratio of a human baseline's action count to the agent's, weights later levels more, and gives an uncleared level zero. We

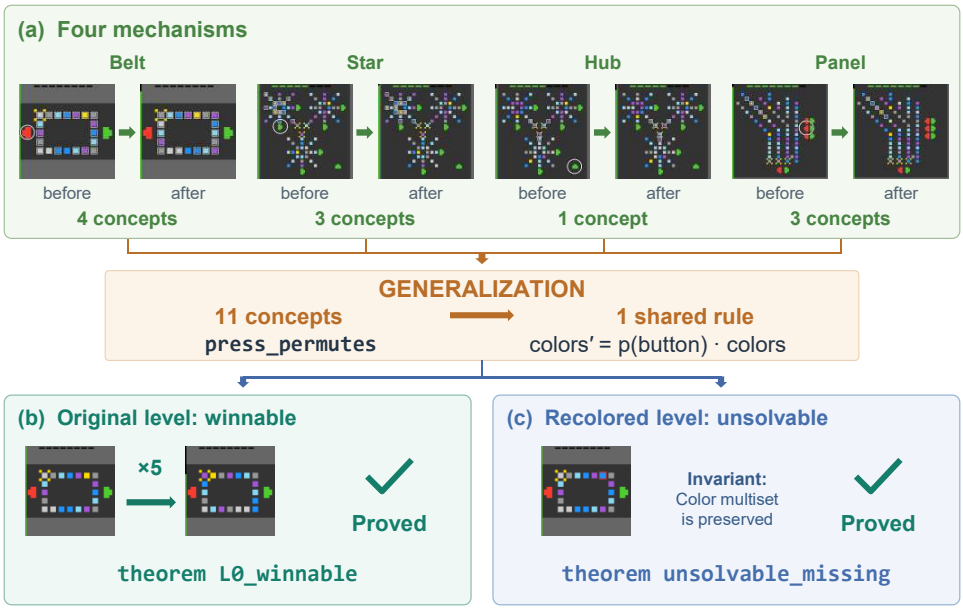


Figure 3: From shared structure to formal proofs in `lp85`. (a) Eleven concepts describing four button mechanisms are generalized into a shared permutation rule. Each before/after pair shows a recorded press, with the pressed button circled and changed tiles outlined. (b) A five-click sequence witnesses reachability. (c) Recoloring the only yellow tile makes the goal unreachable, since every press preserves color counts. Both conclusions are formally proved in Lean 4.

Table 1: Official scores on the 25 public ARC-AGI-3 games. Claude Opus 5 and GPT-6 Astra run under the official standard harness, which provides no code execution; the other systems execute code. AWM's certified clears follow uncertified sessions of the same run in 22 games. The actions for Rodionov's agent are from its ARC Prize scorecard and Memento 3's as reported in (Zhao et al., 2026).

System	Code	Mean score	Games at 100	Certified actions	Actions / human
Claude Opus 5	no	40.7	5	—	—
GPT-6 Astra	no	68.3	10	—	—
GPT-6 Astra (adapter) (ARC Prize Foundation, 2026a)	yes	100.0	25	—	—
Rodionov (2026b)	yes	99.0	23	8,347	0.49
Memento 3 (Zhao et al., 2026)	yes	100.0	25	7,518	0.44
AWM (ours)	yes	99.9	23	**7,448**	**0.43**

Table 2: Formalization and verification of the 25 learned world models in Lean 4. Axioms are those Lean's own logic uses; no proof adds axioms of its own.

Levels replayed	Rules formalized in Lean	Theorems	Axioms
183 / 183	220	1,374	only `propext`, `Classical.choice`, `Quot.sound`

run AWM with Claude Opus 5 on all 25 public games (183 levels), starting each game from scratch. AWM clears every level with a score of 99.9 and the fewest certified actions among the systems reporting this metric in Table 1. We further examine whether the resulting models are compact, reproduce observed behavior in Lean, and generalize beyond the trajectories from which they were learned.

Compact models that reproduce observed behavior. Across the 25 games, AWM records 742 executable concepts and withdraws 143 during model revision. The retained concepts are tested

against the recorded transitions. Structure & Simplify organizes the rest into 170 object types and 283 rules, a median of 6 types and 10 rules per game. Shared patterns include translations, permutations and goal conditions. Lean replays the winning trajectory of every one of the 183 levels from its initial state and verifies its final state against the modeled goal (Table 2).

Generalization beyond observed transitions. We next test the learned rules on behavior the record never showed. We freeze the models of `lp85` and `ft09` after exploration and evaluate them on transitions from random play with previously unseen screens (seeds 0–9; Appendix B.2). The final frozen rules accept every valid transition in the new-trajectory sets. On the two modified `lp85` layout sets, they predict the next tile layout in all 4,160 cases; on the two changed-mechanic `ft09` sets, they reject all 4,031 transitions. On a third game, `ls20`, the same evaluation accepts more than 99.5% of valid transitions on previously unseen screens with the final rules, compared with 67–70% with the concepts before Structure & Simplify (Appendix B.2). Corruption results are reported separately in the supplement.

The changed mechanic modifies `ft09`'s hidden click pattern, causing each click to change 110 or 146 cells instead of 38; rejecting these transitions shows that the rules detect behavior inconsistent with the learned dynamics.

Clearing the games with efficient solutions. The same models are effective for play. AWM with Opus 5 achieves a score of 99.9, compared with 40.7 for Opus 5 under the official standard harness without code execution. It scores 100 on 23 games and at least 97.6 on every game, performing competitively with leading code-executing systems (Table 1). Its certified clears require 7,448 actions, only 43% of the human baseline's 17,135 and the fewest among systems reporting this metric, compared with 7,518 for Memento 3 (Zhao et al., 2026) and 8,347 for Rodionov (2026b). Including exploration and other sessions, AWM uses 24,604 engine actions in total (Appendix B.1). These results show that AWM can construct and use explicit world models while retaining competitive solution efficiency. §4 next examines whether such models also support formal proofs of impossibility.

4 Proving impossibility

A winning trajectory establishes reachability; impossibility requires excluding every legal winning sequence within the budget, including those never explored. We test whether AWM's learned algebraic models enable such exhaustive proofs, and whether those proofs hold for the game itself.

4.1 The game, the seal, and the baselines

In `sc25`, a block moves on a grid and casts spells through a 3×3 panel to reach the exit within the level's budget. Moves and panel clicks cost one unit, completing a spell one more, fuel refunds ten, and the winning move is free. We construct two variants of levels 4 and 5 by rearranging selected elements while preserving the game mechanics, then seal them at budgets of 18 and 35. Exhaustive search on the game engine (the underlying game program), covering all four initial facings and all 13 actions, establishes minimum winning costs of 19 and 36, respectively: neither sealed level can be won, while one additional unit permits a win.

Claude Opus 5.5 and GPT-6 Astra receive the game interface and coding tools, including simulator construction and search; AWM uses Opus 5.5 with the procedure of §2 and a generic Lean skeleton. Each condition has three runs per level, with up to eight reset-delimited attempts per run, ending at game over or recorded abandonment. After each attempt, every agent is asked whether the sealed level can be won and to justify its verdict; AWM constructs Lean proofs under its learned model. Impossibility can be established by proving that the minimum winning cost exceeds the budget. We evaluate proof validity under the learned model separately from numerical agreement with the engine.

4.2 Certified lower bounds

Proving beyond explored trajectories. All six AWM runs produce closed, kernel-checked Lean proofs of impossibility under their learned models and explicit premises, while none of the twelve baseline runs produces such a proof (Table 3). Crucially, these proofs exclude all legal winning

Table 3: Impossibility reasoning across six runs per agent. Bound accuracy is evaluated against the games' minimum winning costs before and after feedback-guided repair. First claim averages the earliest impossibility claim among runs making one, including later withdrawals; it is not proof-completion time. Per-run results are in Appendix C.3.

Agent	Kernel-checked model proofs	Runs claiming impossible	Correct bounds initial → feedback	First claim (attempt)
Claude Opus 5.5	0/6	3/6	1/6 → 1/6	5.3
GPT-6 Astra	0/6	2/6	0/6 → 2/6	8.0
AWM	**6/6**	**6/6**	**4/6 → 6/6**	**2.0**

sequences within the budget under the learned models, including those never explored. For example, an AWM run for level 4 proves

$$\forall B \leq 18,\ \forall \mathbf{a}, \quad \mathrm{Won}(\mathrm{run}_B(w_0, \mathbf{a})) = \mathsf{false}, \tag{4}$$

and verifies a winning witness at $B = 19$. Together, the lower bound and witness certify a tight threshold under the model, whereas finding a winning route alone cannot exclude cheaper alternatives.

AWM establishes such bounds through potential or closure certificates over arbitrary legal action lists. Certificate data may come from any search; Lean checks that the certificate proves the stated bound under the formal model, so the proof need not trust the search implementation. In our mutation tests, the Lean verification package also detects reversed bounds and mismatches between the Python and Lean versions; a separate statement audit reports vacuous premises (Appendix C.7).

Beyond unproved numerical judgments. Only five of twelve baseline runs claim impossibility, four with incorrect bounds. Even the sole correct Claude Opus 5.5 bound lists "that my hand-counted path lengths are truly minimal" as *assumed, not checked*. Finding a winning route establishes an upper bound on the minimum cost, not its optimality. AWM instead provides kernel-checked lower-bound proofs under its learned models.

Figure 4 illustrates this distinction on L4. Astra finds a winning route costing 22 and assumes it is optimal without proof, incorrectly ruling out wins at budgets 19–21. In contrast, AWM formally proves a lower bound of 19 under its learned model and verifies a matching winning witness, establishing a tight bound that agrees with the game's minimum winning cost.

Proof validity and bound accuracy are distinct. Before feedback, four of six AWM bounds agree with the games' minimum winning costs, compared with one for Opus 5.5 and none for GPT-6 Astra.

Some runs identify a correct winning sequence but charge the otherwise free winning move, reporting a bound one unit higher. Because this cost cannot be observed on sealed levels, we count such bounds as correct for all agents only when the discrepancy is explicitly attributable to this accounting convention, not to an incorrect route or cost estimate.

The remaining two AWM bounds, 20 on L4 and 47 on L5, arise from overly restrictive learned rules. Their proofs remain valid under those models, although their numerical thresholds disagree with the game.

4.3 Auditing premises and proving again

After the initial evaluation, the two AWM runs with bounds of 20 and 47 receive feedback only that the game permits a cheaper win, without revealing its cost, route or the faulty rule. AWM revisits its explicit rules, tests alternative formulations through interaction, and constructs new proofs under the refined models (Appendix C.4).

On L4, AWM discovers that growing onto fuel collects it, reducing the bound from 20 to 19. On L5, it identifies terrain rather than limited range as the cause of a failed shot, reducing the bound from 47 to 37 under its convention of charging the winning move, equivalent to the game's minimum cost of 36. Both refinements succeed within four additional attempts, each yielding a new closed Lean proof. With this feedback, AWM reaches correct bounds in all six runs.

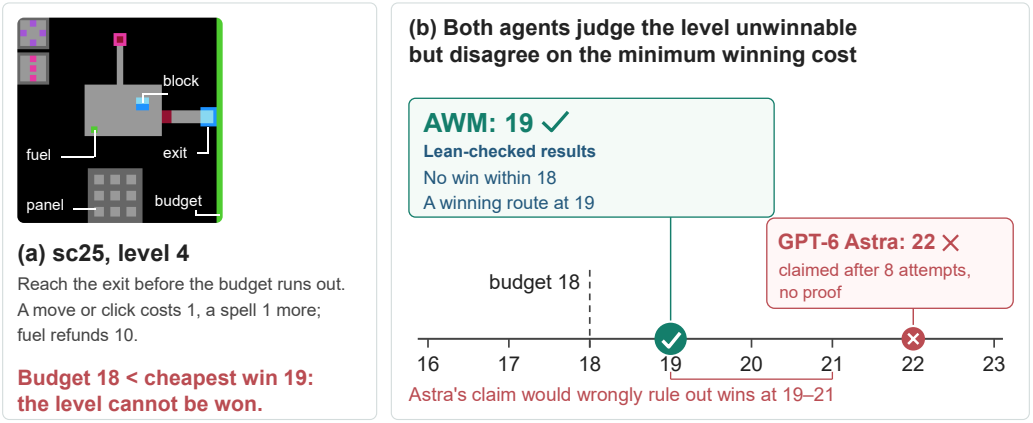


Figure 4: Sealed sc25 level 4. AWM proves in Lean that the minimum winning cost under its learned model is 19, one unit above the budget, and verifies a route at that cost; GPT-6 Astra claims 22 without a proof.

We provide the same discrepancy feedback and additional attempt budget to the four baseline runs with incorrect bounds. Astra corrects both bounds but produces no formal proofs, while Claude Opus 5.5 corrects neither.

The proof points to the error. Both repairs revise explicit rules used by the proofs. A separate injected-error study tests whether proof dependencies help target such checks. On L4, adding these dependencies to per-rule evidence yields a transition exposing the faulty rule in 29 of 42 runs, compared with 16 of 42 given per-rule evidence alone and 30 of 42 told the faulty rule. The dependency-guided condition raises no false alarms in 12 error-free controls (Appendix C.6).

5 Related work

Executable world models. Code world models are learned from interaction for planning (Tang et al., 2024), refined by generation, testing and repair (Dainese et al., 2024) or composed from programmatic experts (Piriyakulkij et al., 2025); theory-based agents explore and plan with learned theories (Tsividis et al., 2021), and TheoryCoder-2 learns reusable abstractions for hierarchical planning (Ahmed et al., 2026). Such models are central to recent ARC-AGI-3 agents (Rodionov, 2026a; Courtis et al., 2026; Rodionov, 2026b): Twin does not act until its simulator reproduces every observed transition (Skoutnev et al., 2026), and Tycho finds that a closer match does not by itself yield better play (Lehmann et al., 2026). The closest work, Memento 3, compiles a natural-language rulebook into code and accepts an update only if an LLM judges it faithful and replay is exact (Zhao et al., 2026). Yet a code world model can pass a sampled-transition check at 100% accuracy and still lose because it misses a key rule (Aguilar Martín, 2026). AWM instead abstracts its rules into a shared structure, tests them beyond the record, and proves reachability and impossibility, checking the proofs' premises against the game.

Certified impossibility. Unsolvability certificates let an independent checker confirm that a planning task has no plan (Eriksson et al., 2017), in the spirit of certifying algorithms (McConnell et al., 2011), and certificate checking has been formally verified for temporal planning (Wang & Abdulaziz, 2025); all start from a supplied model. Impossibility transfers from an abstraction only if it over-approximates the real system, budget semantics included (Clarke et al., 1994), and action-model learning gives conditions under which learned models do so (Aineto & Scala, 2024). AWM learns its model from interaction; Lean checks every deduction from that model, and interaction checks the model's rules against the game.

Formal reasoning. Neural provers produce Lean proofs at olympiad level (Achim et al., 2025; Chen et al., 2025; Hubert et al., 2026; Ren et al., 2025), AI systems have produced Lean proofs of open Erdős problems (Sothanaphan, 2026; Tsoukalas et al., 2026), and autoformalization translates informal

statements into formal ones (Wu et al., 2022). There the objects of proof are given mathematics; AWM must first infer the dynamics and costs its proofs quantify over, which makes the fidelity of its formal model an empirical question. Large-scale AI-assisted software verification proves properties of compiled programs relative to machine semantics and theorem statements, and can expose bugs during proof construction (Lucas et al., 2026). That work also studies Claude-generated x86 semantics; AWM learns the objects, action rules and goals of an unfamiliar interactive environment.

6 Limitations and conclusion

Limitations. We demonstrate AWM on ARC-like environments with discrete actions and immediate, observable feedback, where learned rules can be checked against recorded transitions; richer hidden state, delayed feedback and stochastic dynamics remain to be tested. Lean guarantees the correctness of proofs under the learned model's explicit premises, and whether these premises capture the environment is tested by interaction: the explicit rules and kernel-checked proofs separate errors in the learned model from errors in reasoning, so a discrepancy can be traced to its assumption and addressed by further exploration. Broader environments and tasks are the natural next test of such formal reasoning.

Conclusion. Formal reasoning traditionally begins with predefined objects and rules. AWM takes a step toward reasoning rigorously about unknown worlds by discovering these structures through interaction, organizing them into compact algebraic models, and formalizing them in Lean. The resulting models make learned mechanisms explicit and their behavior checkable step by step, while supporting kernel-checked proofs of reachability and impossibility, including claims about action sequences never explored. By making the assumptions behind these proofs explicit, AWM also allows discrepancies with the environment to be traced to learned rules and addressed through further exploration. Together, these capabilities bridge empirical exploration and formal reasoning, extending mathematical proof beyond problems whose definitions are given in advance.

Reproducibility statement

The readings of §2 and §3 come from the benchmark repository at commit `fa928185` (built 2026-10-24). Each game's model, tests and Lean project are regenerated by one command per game; the per-game Lean projects are checked with Lean 4.34.0 and do not use Mathlib. The readings of §4 come from the archive of the budget-sealed instance: both engine versions with their md5, the abstract model and its cross-check scripts with their raw outputs, the Lean files with their compile logs and axiom audits, every file each agent wrote, and the verbatim end-of-attempt interviews; the per-run readings are given in the appendix. The game environments are third party; official replays are linked rather than redistributed.

Statement on AI use

The agents studied in this paper are LLM agents, so AI use is part of the method. Generative AI was also used to draft and edit the text and to write the analysis scripts. All reported numbers come from script outputs over the recorded runs, not from model-generated text. The Lean proofs are checked by the Lean kernel, independently of any model.

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A Method details

This appendix gives the definitions behind §2: concepts and the gate (§2.2), the simplification test (§2.3), and the Lean statements, certificates and their transfer to the environment (§2.4).

A.1 Environment, concepts and the gate. We model an environment as a deterministic, goal-directed special case of a partially observable Markov decision process (Kaelbling et al., 1998): states $\mathcal{S}$, actions $\mathcal{A}$, observations $\mathcal{O}$, a transition function $T : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$, an observation function $\rho : \mathcal{S} \to \mathcal{O}$ and goal states $G \subseteq \mathcal{S}$. The agent sees $o_t = \rho(s_t)$ and receives a signal when a level is cleared. A transition is a triple $x = (o, a, o')$, and a game's transition corpus $\mathcal{D}$ is the set of distinct transitions across its runs, deduplicated by content hash.

A concept $c = (\sigma_c, \alpha_c, \gamma_c, E_c)$ has an optional executable scope σ_c, an assertion α_c returning TRUE, FALSE or NA, an optional predictor γ_c, and a non-empty evidence set E_c. When the scope rejects x the verdict $v_c(x)$ is NA, otherwise it is $\alpha_c(x)$; an exception, an invalid return value or a timeout gives ERR. A concept is scored only on transitions outside E_c. The gate of Eq. (1) is enforced on the action command: it refuses the next action while the latest settlement has a FALSE or ERR verdict on any $x \in \mathcal{D}$, including the concept's own evidence. Figure 5 shows one concept.

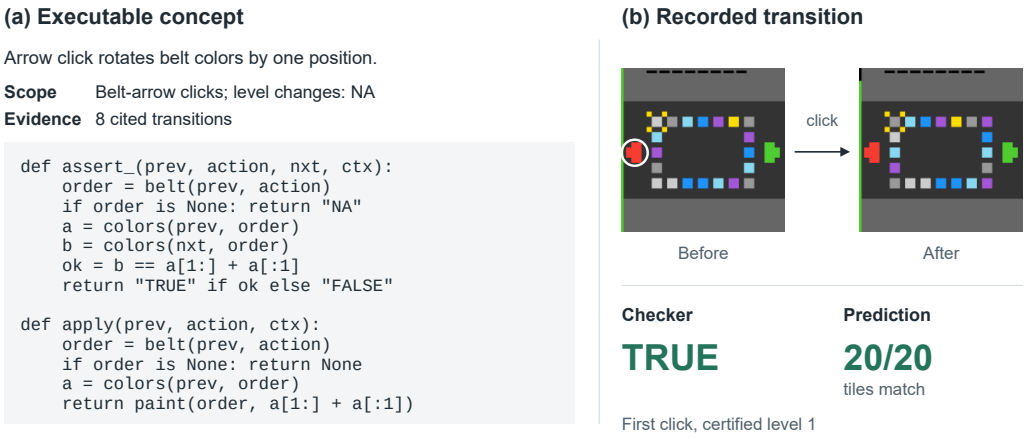


Figure 5: An executable concept in `lp85`. (a) Scope, evidence and abridged code for a click on an arrow pointing against its belt's order; helper names are simplified. (b) On the first click of the certified level-1 run, the checker returns `TRUE` and the prediction matches all 20 tiles.

A.2 Objects, rules and the simplification test. A partial parser $\pi : \widetilde{\mathcal{O}} \rightharpoonup \mathcal{W}$ maps an observation, augmented by any explicitly supplied history or auxiliary information, to a world of typed objects; $\pi(o)$ abbreviates parsing with that context. Pixel interpretation is confined to the parser, and parse failures are reported rather than guessed. A rule $r = (\sigma_r, \alpha_r)$ has a scope and an assertion on $(\pi(o), a, \pi(o'))$. The diagnostic set $\mathcal{N}$ of Eq. (2) contains at most 120 corrupted transitions per game, with a fixed seed. Each replaces part of a recorded transition (o, a, o') in one of five ways: the successor is o itself; the successor of the same action on another transition; the predecessor of another transition; o' with a few cells changed that do vary in the corpus; or another transition's o. We set

$$\mathrm{Rej}(r) = \{m \in \mathcal{N} : v_r(m) = \mathsf{FALSE}\}. \tag{A.1}$$

The following is an auxiliary diagnostic within the four transformations, not a separate operation or a sufficient deletion criterion: rule r has no unique rejection on $\mathcal{N}$ when

$$\mathrm{Rej}(r) \subseteq \bigcup_{r' \neq r} \mathrm{Rej}(r'). \tag{A.2}$$

Any removal must preserve model behavior and pass the same before/after checks; an analogous field diagnostic fixes a field to a constant and compares verdicts on $\mathcal{D} \cup \mathcal{N}$. Figure 6 shows the test on `lp85`.

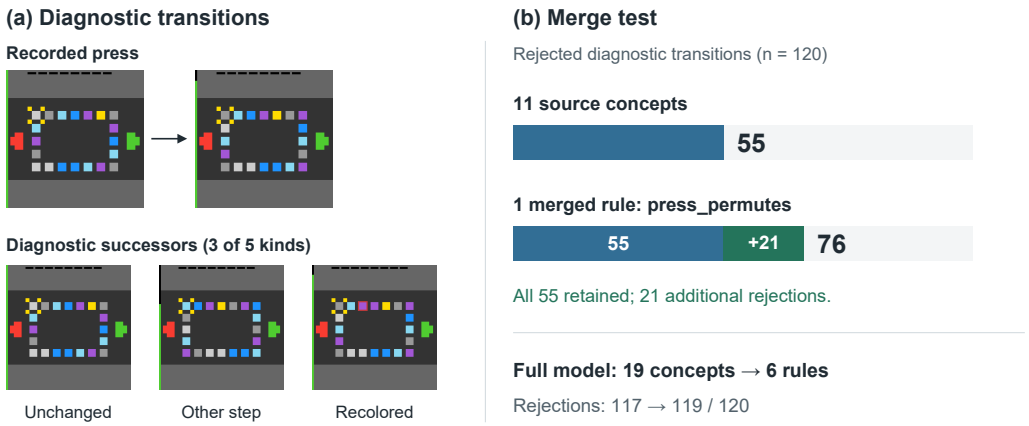


Figure 6: Testing a merge on `lp85`. (a) A recorded press and three of the five diagnostic constructions. (b) On the same 120 diagnostic transitions, `press_permutes` preserves all 55 rejections of its eleven source concepts and adds 21. Lack of unique rejections alone does not justify deletion: `dead_click`, `board_static` and `bar_never_drops` are retained based on further tests.

A.3 Lean encoding, certificates and transfer. Formalization starts from the consolidated objects and rules. AWM encodes object types as Lean `structure` or `inductive` declarations and primitive rules as executable `def` definitions; consequences derivable from these definitions are proved as `theorem` statements in the derived layer. The agent assembles the modeled transition and goal rules in `Step.lean`, whose `step` function dispatches to the named rule definitions and returns a successor or a terminal outcome for actions admitted by the model's legality conditions. For example, in `lp85`, a click either leaves the colors unchanged or applies a button's permutation, after which the goal rule determines whether the level is won; color conservation is a theorem derived from these definitions. This shared transition function is used both to replay recorded trajectories and to reason about unobserved action sequences.

On this basis, $\mathrm{everWon}(w, \bar{a})$ holds when some legal prefix of $\bar{a}$ reaches a winning outcome. The two statements of §2.4 are

$$\mathrm{Winnable}(w) :\Leftrightarrow \exists\, \bar{a}.\ \mathrm{everWon}(w, \bar{a}) = \mathsf{true}, \qquad \mathrm{Unsolvable}(w) :\Leftrightarrow \forall\, \bar{a}.\ \mathrm{everWon}(w, \bar{a}) = \mathsf{false}.$$

For level ℓ, the initial world w_0^ℓ is parsed from that level's first screen in the certified run, and the certified actions are $\bar{a}^\ell$. The reachability proof establishes $\mathrm{traceOk}(w_0^\ell, \bar{a}^\ell)$, i.e. that every computed state agrees with the recorded frame on the components the formal model represents, and $\mathrm{Winnable}(w_0^\ell)$ with $\bar{a}^\ell$ as witness. An impossibility proof establishes $\mathrm{Unsolvable}$ within a budget through a potential satisfying Eq. (3). Each theorem is closed by kernel reduction, its axioms lie within $\{$`propext`, `Classical.choice`, `Quot.sound`$\}$, and no proof uses `native_decide`; compiled with Lean 4.34.0, the published `Game.lean` of every game compiles and its `consistent` and `faithful` theorems depend only on `propext`.

The conclusions transfer to the environment under the following condition (Aineto & Scala, 2024; Clarke et al., 1994). Let $h : \mathcal{S} \to \mathcal{W}$ map concrete states, augmented with observation history where required, to modeled states.

Proposition A.1. If the model $\widehat{\mathcal{M}}$ over-approximates the environment, with matching initialization, every relevant real transition represented, real goals mapped to model goals, and cost and budget semantics that do not exclude a cheaper real execution, then

$$\mathrm{Unsolvable}_{\widehat{\mathcal{M}}}\big(h(s_0)\big) \implies \forall\, \bar{a}.\ T^{\bar{a}}(s_0) \notin G.$$

Agreement with recorded transitions establishes this premise only on the transitions it covers, so AWM states its impossibility results as $\mathrm{Unsolvable}$ of the learned model.

Proof. Suppose a legal real execution reaches a goal within budget. By matching initialization and transition simulation, induction on its prefixes gives a corresponding legal model execution. Goal preservation and the cost condition make this a model win within budget, contradicting $\mathrm{Unsolvable}_{\widehat{\mathcal{M}}}(h(s_0))$. □

B TWENTY-FIVE GAMES

This appendix gives the per-game data behind §3, Table 1 and Table 2, the test of frozen rules on unseen transitions, and the `lp85` proof sizes of §2.5.

B.1 Per-game results. Table 4 gives AWM's results and model for each game; Table 5 gives the official scores of all systems.

Table 4: AWM on each game. *Score*: official score of the certified run. *Cert.*: actions in the certified clears, which the score counts. *Sent*: every action AWM sent to the engine, including uncertified sessions, failed attempts and resets. *Human*: actions of the human baseline. *Concepts*: concepts written during play, with those withdrawn in brackets. *Rules (Lean)*: rules translated into Lean. *Replayed*: levels whose certified winning run Lean replays from the first screen to the modeled goal.

Game	Lv.	Score	Cert.	Sent	Human	Concepts	Obj. types	Rules	Rules (Lean)	Replayed
ar25	8	100.0	253	253	748	17 (3)	6	8	5	8/8
bp35	9	100.0	374	374	651	26 (4)	3	11	10	9/9
cd82	6	100.0	70	661	171	44 (13)	7	11	10	6/6
cn04	6	100.0	261	745	789	22 (1)	4	9	8	6/6
dc22	6	100.0	1,037	2,664	1,228	48 (6)	8	17	14	6/6
ft09	6	100.0	80	168	208	15 (6)	6	7	4	6/6
g50t	7	100.0	288	811	879	20 (1)	10	13	11	7/7
ka59	7	97.6	421	4,061	730	22 (8)	9	11	10	7/7
lf52	10	100.0	905	905	1,339	47 (6)	12	11	10	10/10
lp85	8	100.0	79	200	388	23 (4)	6	6	4	8/8
ls20	7	100.0	368	1,467	776	22 (3)	12	21	18	7/7
m0r0	6	100.0	215	486	1,107	26 (5)	8	9	8	6/6
r11l	6	100.0	81	184	233	26 (4)	6	9	8	6/6
re86	8	100.0	547	2,153	1,255	40 (4)	7	12	8	8/8
s5i5	8	99.3	461	2,618	638	49 (9)	5	19	17	8/8
sb26	8	100.0	124	424	213	20 (1)	6	16	12	8/8
sc25	6	100.0	146	401	350	27 (1)	5	18	11	6/6
sk48	8	100.0	263	1,407	1,070	39 (8)	6	7	4	8/8
sp80	6	100.0	138	371	518	58 (23)	3	10	6	6/6
su15	9	100.0	90	904	361	37 (15)	5	16	12	9/9
tn36	7	100.0	125	292	317	33 (9)	5	7	4	7/7
tr87	6	100.0	119	430	414	13 (3)	6	7	4	6/6
tu93	9	100.0	187	566	462	17 (0)	8	10	8	9/9
vc33	7	100.0	167	538	447	14 (0)	9	10	7	7/7
wa30	9	100.0	649	1,521	1,843	37 (6)	8	8	7	9/9
Total	183	99.9	7,448	24,604	17,135	742 (143)	170	283	220	183/183

B.2 Frozen rules on unseen transitions. We evaluate whether learned rules generalize to unseen screens and detect changes in game mechanics. For `lp85` and `ft09`, we freeze the concepts recorded during play (19 and 9, respectively) and the rules after Structure & Simplify (6 and 7). We collect transitions from random presses (seeds 0–9), retaining only states whose screens never appeared in the recorded gameplay.

Each game is evaluated on new trajectories from the original levels (T1) and two modified levels, whose winnability is verified by exhaustive search. In `lp85`, the modifications preserve the mechanics but change the layout: U moves one goal tile, making level 6 unwinnable, while D adds another swap that restores winnability. In `ft09`, the modifications change the hidden click mechanic: FU replaces level 2's click pattern with a plus shape, making it unwinnable, while FD adds a one-pixel change that restores winnability.

A rule set accepts a transition if no rule judges it false, and rejects it if at least one rule judges it false. We also generate invalid transitions using five corruption operators to test whether the rules can detect inconsistencies. Table 6 reports the results before and after Structure & Simplify. The frozen rules generalize to modified layouts in `lp85`, while detecting all tested transitions as inconsistent after the mechanic changes in `ft09`.

`ls20`. On `ls20`, the concepts written during play accept 67–70% of valid transitions on unseen screens; after Structure & Simplify, the frozen rules accept 99.47–99.95% and reject 99.1–99.3% of corrupted transitions, against 71.8–100% for the concepts (Table 7). The gap comes from wall bumps: the concepts `energy_tick` and `energy_cost` charge energy for a blocked move only on the

Table 5: Official scores on the 25 public ARC-AGI-3 games. Baselines are the official ARC Prize results: Claude Opus 5 (High) and GPT-6 Astra (Max) under the standard harness, which provides no code execution, and GPT-6 Astra (Max) under its provider-adapter harness. AWM is scored on its certified run. Rodionov's agent from its ARC Prize scorecard (Rodionov, 2026b); Memento 3 as reported in (Zhao et al., 2026).

		Official score (max 100)					
Game	Lv.	Claude Opus 5	GPT-6 Astra	GPT-6 Astra (adapter)	Rodionov (2026b)	Memento 3 (Zhao et al., 2026)	**AWM (ours)**
ar25	8	100.0	100.0	100.0	100.0	100.0	**100.0**
bp35	9	0.0	2.2	100.0	100.0	100.0	**100.0**
cd82	6	0.6	82.9	100.0	100.0	100.0	**100.0**
cn04	6	47.6	100.0	100.0	100.0	100.0	**100.0**
dc22	6	44.8	100.0	100.0	100.0	100.0	**100.0**
ft09	6	100.0	100.0	100.0	100.0	100.0	**100.0**
g50t	7	0.0	0.0	100.0	100.0	100.0	**100.0**
ka59	7	10.4	75.0	100.0	100.0	100.0	**97.6**
lf52	10	1.8	10.9	100.0	100.0	100.0	**100.0**
lp85	8	100.0	100.0	100.0	100.0	100.0	**100.0**
ls20	7	5.4	83.5	100.0	100.0	100.0	**100.0**
m0r0	6	28.6	71.4	100.0	100.0	100.0	**100.0**
r11l	6	100.0	100.0	100.0	100.0	100.0	**100.0**
re86	8	58.3	77.8	100.0	100.0	100.0	**100.0**
s5i5	8	100.0	100.0	100.0	100.0	100.0	**99.3**
sb26	8	77.8	100.0	100.0	100.0	100.0	**100.0**
sc25	6	0.0	47.6	100.0	84.2	100.0	**100.0**
sk48	8	8.3	19.7	100.0	100.0	100.0	**100.0**
sp80	6	56.3	100.0	100.0	100.0	100.0	**100.0**
su15	9	10.4	27.4	100.0	100.0	100.0	**100.0**
tn36	7	6.0	53.6	100.0	90.1	100.0	**100.0**
tr87	6	47.6	100.0	100.0	100.0	100.0	**100.0**
tu93	9	3.6	3.1	100.0	100.0	100.0	**100.0**
vc33	7	98.8	73.3	100.0	100.0	100.0	**100.0**
wa30	9	10.6	80.0	100.0	100.0	100.0	**100.0**
Mean		40.7	68.3	100.0	99.0	100.0	**99.9**
Games at 100		5	10	25	23	25	**23**

Table 6: Frozen-rule evaluation on unseen transitions, before / after Structure & Simplify (%). lp85 U/D test generalization to modified layouts, while ft09 FU/FD test detection of changed mechanics. "Wrong rejected" reports the range over the four corruption operators for which both rule sets can read both frames. The 0% acceptance on FU/FD indicates that all tested transitions violate the frozen rules: the "nothing else changes" rule detects clicks changing 110 or 146 cells instead of 38.

Set	n	Transitions accepted	Wrong rejected	Next frame predicted
lp85 T1 (new trajectories)	2,073	100 / 100	99.5–100 / 99.5–100	99.95 / 99.95
lp85 tile swap U	2,080	100 / 100	100 / 100	100 / 100
lp85 tile swap D	2,080	100 / 100	100 / 100	100 / 100
ft09 T1 (new trajectories)	2,026	99.0 / 100	99.0–100 / 100	—
ft09 pattern change FU	2,015	0 / 0	—	—
ft09 pattern change FD	2,016	0 / 0	—	—

two levels where play recorded one, while the rule `energy_cost` charges every wall bump the level's step cost, which holds on screens the agent never saw.

Table 7: Frozen concepts and rules on two `ls20` test sets of previously unseen screens (%). T1: new trajectories on the original levels; T2V: level 1 with its switch moved one cell. Corrupted rejected: range over the four corruption operators for which both sides can read both frames.

		Valid accepted		Corrupted rejected	
Test set	n	Concepts	Rules	Concepts	Rules
T1	2,017	67.0	99.95	75.3–100	99.1–99.3
T2V	2,074	70.3	99.47	71.8–100	99.1–99.2

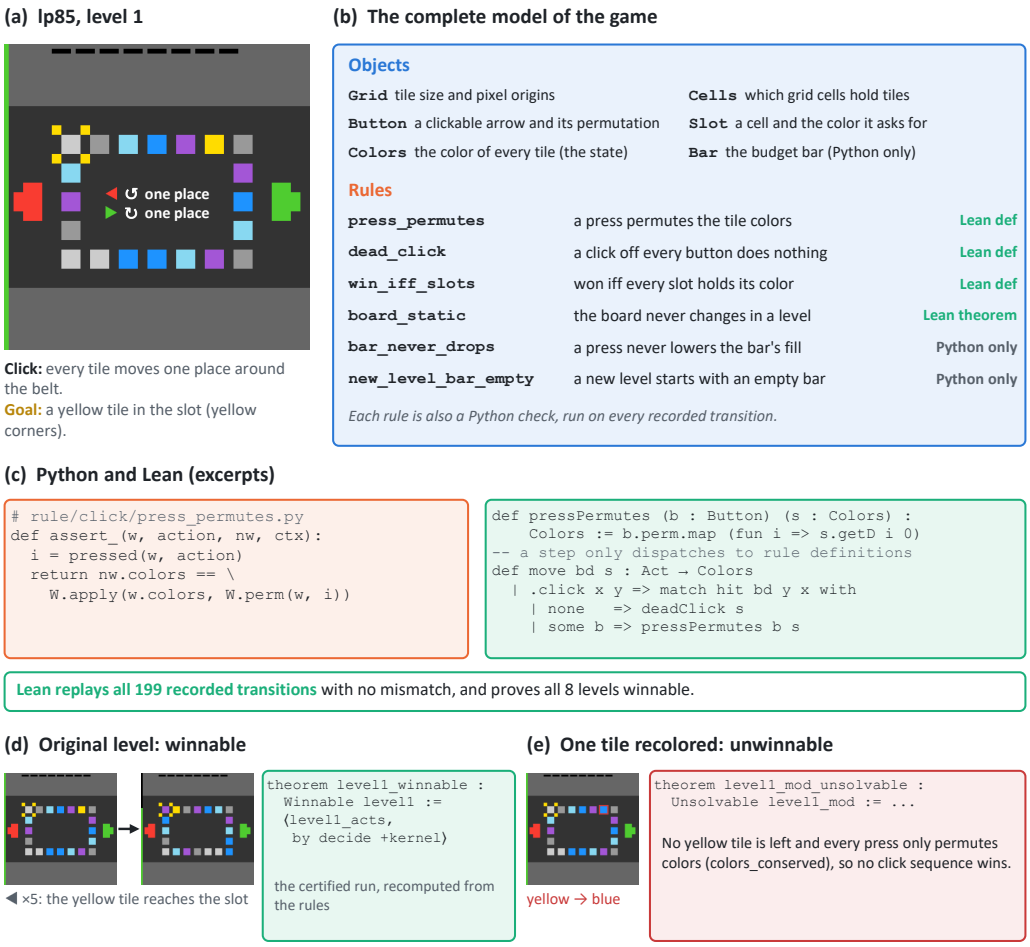


Figure 7: The complete algebraic world model of an ARC-AGI-3 game. (a) Level 1 of `lp85`: 20 colored tiles form a closed belt, and each click on an arrow moves every tile one place around the belt, counterclockwise for the left arrow and clockwise for the right; the level is won when a yellow tile stands in the slot marked by yellow corners. (b) All objects and rules of the model, with the form each takes in Lean; the two rules about the budget bar stay in Python. (c) Excerpts: one rule as its Python check, the same rule as a Lean definition, and the Lean step, which only dispatches to rule definitions. Lean replays every recorded transition of the game with no mismatch. (d) On the original level, Lean checks a winning witness, the certified run. (e) With the only yellow tile recolored, the conservation law of `press_permutes` shows that no click sequence wins. Both statements are theorems about the model.

B.3 `lp85`: the complete model and proof size. Figure 7 shows the complete model of `lp85`. Claude Opus 5.5 proves each of 34 `lp85` tasks winnable (with a winning click sequence) or unwinnable (with a Lean proof), once with model A, the eleven concepts written during play, and once with model B, the single `press_permutes` rule; three runs per task and model, 204 runs in all, every verdict checked against exhaustive search on the engine. Table 8 gives the results.

Table 8: Proving 34 `lp85` tasks with the eleven concepts (A) and the merged rule (B): runs proved, and medians of lemmas, proof lines and kernel time over the runs of each group. No run gives a wrong answer; A's three unproved runs are two timeouts and one run that stopped without an answer. On winnable instances both models are proved by a click sequence and `decide`.

Instances	Runs	Proved A / B	Lemmas A / B	Lines A / B	Kernel (s) A / B
original levels (winnable)	24	24 / 24	0 / 0	4.5 / 4	1.00 / 0.83
goal colour removed (unwinnable)	24	24 / 24	6 / 3	75.5 / 28	0.90 / 0.80
level 6 swaps (one each)	6	6 / 6	2 / 1.5	36.5 / 24	1.69 / 1.22
large layouts (winnable)	18	16 / 18	0 / 0	5.5 / 4	1.95 / 1.00
large layouts, altered (unwinnable)	18	17 / 18	12 / 4	117 / 49.5	2.82 / 2.41
ring spacing (unwinnable)	12	12 / 12	12 / 6.5	111 / 62.5	4.27 / 3.02
all	102	99 / 102			

B.4 Checking a model-level proof against the engine. This check concerns the level-6 tile-swap variant U of Appendix B.2. We export its engine-derived, budget-independent reachable configuration graph as a Lean table, with 7,500 configurations and 52,500 transitions for the seven nontrivial button actions. A Lean proof checks each transition against the premises of the model-level unsolvability theorem, so the conclusion transfers to the exported graph. The proof uses only Lean's standard axioms. Extracting the graph from the game engine remains outside the Lean kernel's guarantee.

C Sealed levels

This appendix gives the construction, protocol and per-run results behind §4 and Table 3, and the experiments of §4.2.

C.1 Construction and engine ground truth. We locate the field that stores the budget and confirm that editing only it changes only the budget. We find $B^\star$, the cost of the cheapest win, by exhaustive search on the engine, and seal the level at $B^\star - 1$. The search is completed before any run starts. The search runs on the sealed engine itself, over all four opening facings and all 13 actions (four moves and nine panel cells) from every reachable configuration, 905,587 on level 4 and 1,423,609 on level 5. The counter is kept outside the state key, at its least value per configuration; the search covers all six writes to the counter in the source and its two checks against the cap. Clicks outside the nine panel cells, scanned over the whole 64×64 screen on 30,001 sampled frame classes per level, change none of the 13 actions' outcomes. Table 9 gives the result, and Figure 8 shows level 4.

Table 9: The two sealed levels. Costs are in counter units; "gross" adds back the fuel refund of 10. At the sealed cap no state wins from any opening; the optimal route wins on the unmodified engine at cap $B^\star$ and loses at $B^\star - 1$.

Level	Public cap	Sealed cap	$B^\star$ (gross)	Optimal route
4	35	18	19 (29)	27 actions
5	65	35	36 (46)	42 actions

C.2 Agents and protocol. Three configurations play each sealed level, three runs each. AWM is Claude Opus 5.5 with the AWM procedure, a generic Lean skeleton and tools for recording predictions and checking rules, but no game rules. Claude Opus 5.5 and GPT-6 Astra without AWM receive the task, the game interface and the ordinary tools of a coding agent. Each run has up to eight attempts on the sealed level, and an attempt ends at a game over. After each attempt the agent answers the

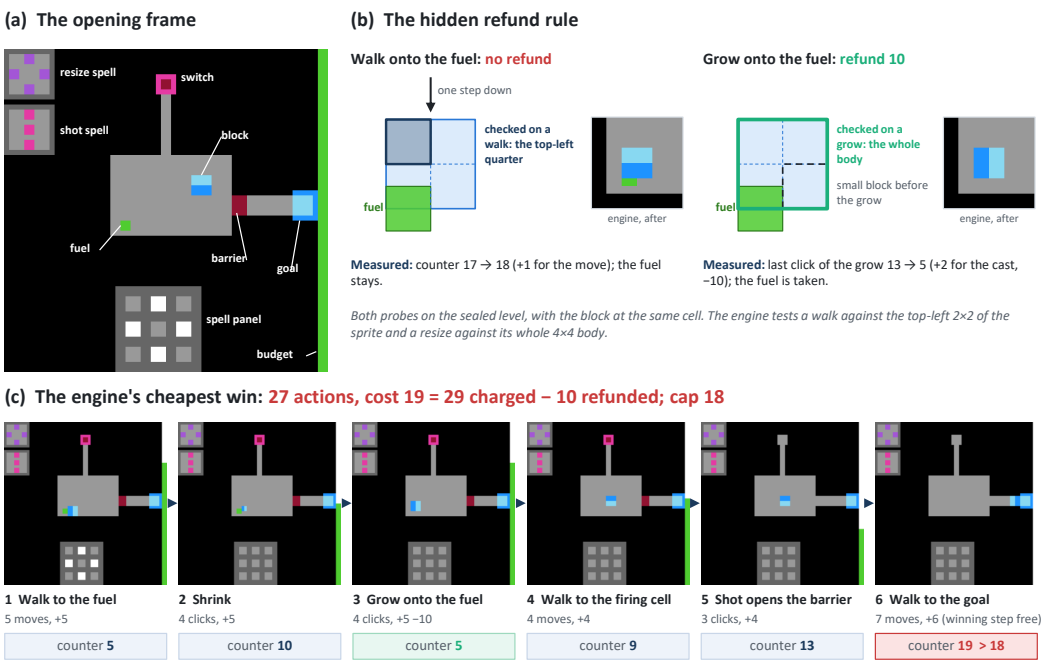


Figure 8: Sealed level 4 of `sc25`. (a) The opening frame. (b) The hidden refund rule: with the fuel under the bottom-left quarter of a large block, walking there pays nothing, because the engine checks only the top-left quarter, while growing there refunds 10; this is the rule the AWM run that proved 20 misread. (c) The engine's cheapest win, 27 actions at a cost of 19 against a cap of 18, with the counter after each stage. Frames are rendered from the real engine with the official palette; those in (c) are replayed on an otherwise identical level whose cap is 19, since on the sealed level the 26th action ends the game.

interview of Appendix D, which records a verdict and its argument. Arenas carry opaque names, contain no ground-truth file, and start with cross-run memory cleared.

C.3 Per-run results. Table 10 lists the 18 runs of Table 3.

Table 10: The 18 runs of Table 3. *First claim*: the attempt of the first impossibility verdict. *Bound*: the least winning cost the run reports before feedback, in its own units for runs without AWM; $B^\star$ is 19 on level 4 and 36 on level 5. A bound is correct if it equals $B^\star$ or exceeds it only by charging the winning move, which the engine does not charge (37 on level 5; 42 actions against a budget of 41 in `fbac96`). *Lean proof*: a closed, kernel-checked impossibility theorem under the run's model.

Level	Agent	Run	First claim	Final verdict	Bound	Correct	Lean proof
4	AWM	`9ec8d4`	2	impossible	20	no	yes
4	AWM	`69fe87`	1	impossible	19	yes	yes
4	AWM	`2a3cc5`	2	impossible	19	yes	yes
4	Claude Opus 5.5	`2f1a1c`	—	undecided	—	no	no
4	Claude Opus 5.5	`d267df`	—	undecided	—	no	no
4	Claude Opus 5.5	`1dd6e6`	—	undecided	—	no	no
4	GPT-6 Astra	`44ec53`	8	undecided	22	no	no
4	GPT-6 Astra	`4d3b31`	8	impossible	22	no	no
4	GPT-6 Astra	`98893b`	—	undecided	—	no	no
5	AWM	`284967`	3	impossible	37	yes	yes
5	AWM	`4f4628`	2	impossible	36	yes	yes
5	AWM	`977714`	2	impossible	47	no	yes
5	Claude Opus 5.5	`339b0a`	4	impossible	47–49	no	no
5	Claude Opus 5.5	`202da3`	6	impossible	43	no	no
5	Claude Opus 5.5	`fbac96`	6	impossible	42 (budget 41)	yes	no
5	GPT-6 Astra	`2cdca5`	—	undecided	—	no	no
5	GPT-6 Astra	`3eb226`	—	undecided	—	no	no
5	GPT-6 Astra	`b3830e`	—	undecided	—	no	no

C.4 Repair. After the eight attempts, the two AWM runs and the four runs without AWM whose bounds are wrong each receive one round of feedback: the engine has a cheaper win than the run's analysis allows. No rule, cost or route is named. The AWM message adds a self-audit instruction (relax rules consistently with the record, test the disagreements in the game, revise, prove again); the message to runs without AWM states the same logic in plain language. Each run gets at most four further attempts. Table 11 gives the results.

Table 11: The repair round of §4.3. Bounds of runs without AWM are in their own units, with the budget they state.

Run	Level	Attempts	Rule revised	Bound after	Lean proof
AWM `9ec8d4`	4	4	growing onto fuel collects it	19	yes
AWM `977714`	5	4	terrain, not range, stops the shot	37	yes
GPT-6 Astra `44ec53`, `4d3b31`	4	$\leq$4	—	19 (both)	no
Claude Opus 5.5 `339b0a`, `202da3`	5	$\leq$4	—	42 (budget $\sim$40), 68 (budget 64)	no

C.5 Transfer to the engine. We repair a level-5 model learned by AWM by replacing each transition the engine refutes with the observed one and charging the winning move as the engine does. The repaired model gives a bound of 36, and its certificate holds on every engine transition within cap 35 from the opening that `set_level` produces: 17,194,487 non-terminal engine transitions, evaluated as 32,523 distinct abstract queries, with no violation.

C.6 Locating an error from the proof. We inject an error that changes a single rule into the reference model of each sealed level, which matches the engine under the abstractions of §C.1; the

main sample is the 14 injections on level 4 and 7 on level 5 that change the model's least winning cost, each run with seeds 1–3. Claude Opus 5.5 receives the faulty model with its proof and a probe service on the sealed engine, and is asked to find where the model and the engine disagree; C1 also receives a per-rule evidence table, C2 the same table and the proof's rule dependencies without the faulty rule named, and R is told which rule is faulty, while C0 probes at random without an agent and EX searches the engine exhaustively. A discovery is an observation in the scope of the faulty rule on which the engine disagrees with the model, counted in engine actions up to the first one, within a budget of 50 actions on level 4 and 5,000 on level 5.

Table 12: Locating an injected error (14 injections on level 4, 7 on level 5, three seeds each). *Found*: discovery within the action budget. *Median*: Kaplan–Meier median of actions to the first discovery, censored at the budget. *Named*: runs whose conflict log or final model points to the faulty rule. EX has no budget. The paired difference in discovery between C2 and C1 over the 63 sample–seed pairs is 0.238 (sample-clustered bootstrap 95% interval $[0.079, 0.397]$). On 12 error-free control runs C2 raises 0.00 false alarms per run and C1 0.33.

Arm	L4 found	L4 median	L5 found	L5 median	Found (L4 + L5)	Named	Tokens (median)
C0 random	12/42	> 50	0/21	> 5000	12/63	—	—
C1 per-rule evidence	16/42	> 50	1/21	> 5000	17/63	35/63	2.35M
C2 proof dependencies	29/42	27	3/21	> 5000	32/63	50/63	3.29M
R faulty rule named	30/42	26	5/21	> 5000	35/63	61/63	1.98M
EX exhaustive	14/14	136,852	1/7	—	15/21	—	—

A level-5 comparison on 42 paired runs per arm finds the faulty rule within the 5,000-action budget in 8 of 42 runs for C2 and 5 of 42 for C1 (Table 13).

Table 13: Level-5 injected-error comparison on 42 runs per arm, using the discovery criterion of Table 12.

Arm	L5 found	Discovery rate
C1 per-rule evidence	5/42	11.9%
C2 proof dependencies	8/42	19.0%

C.7 Lean against a Python checker. On three models, the reference models of the two sealed levels (2,675 and 3,502 states) and the merged `1p85` model (869 states), we compare a Python search for the least winning budget (V1), a Python checker of the certificate conditions (V2) and the Lean verification package (V3), with 21 mutants per model: errors in the certificate generator (G), in the model definition (M1–M6), a change to only one of the Python and Lean versions (M7), and errors in the statement (S1–S7). V3 includes a separate statement audit that reports vacuous premises or an empty start set; the main theorem can still compile, since the kernel checks whether a conclusion follows from its stated premises. Table 14 gives the results.

Table 14: Mutation test of three routes to the same bound, three models together. V3 includes Lean proof checks and the separate statement audit; reporting a vacuous premise is not rejection by the Lean kernel. n/a: the route does not read the object that the mutant changes.

Mutant	V1 Python search	V2 Python checker	V3 Lean + audit
generator error: invalid certificate rejected	n/a	12/12	12/12
of which the conclusion is false on the engine	n/a	9/9	9/9
model error (M1–M6) rejected	none	none	none
Python and Lean versions differ (M7) rejected	0/3	0/3	3/3
vacuous premises or empty start set (S1, S2) reported	0/6	0/6	6/6
reversed bound (S4) rejected	n/a	0/3	3/3
wrong goal set, other model version (S3, S7) rejected	none	none	none

D Interview prompts

This appendix reproduces verbatim the interview of Appendix C.2. After each attempt the harness sends questions 1–3, the formalization questions (AWM runs only) and the verdict lines, in the same session as the play; the level is then reset with the retry message. After the last attempt the final interview replaces questions 1–3 and keeps the same verdict lines.

Questions 1–3, every run.

```
That attempt failed. The game will reset this level, and you may get another attempt.

Answer the following. Keep it to what it needs; do not pad.

1. What happened on the level you were on when the attempt ended, and what do you think made
   it fail? Explain in plain prose, as if to someone who did not watch you play.

2. Do you still see a way to beat this level?
   - If you do: say what it is, and what you would still need to check to be sure of it.
   - If you no longer do: say why not. State precisely what you are claiming, give the
     argument for it, and make clear what the argument rests on - which parts you established
     and which are assumptions you did not check. Someone else should be able to follow it and
     judge whether it holds. The form of the argument is yours to choose.
   - If you cannot tell either way, say so, and say what would settle it.

3. What will you do on the next attempt?
```

Formalization questions, AWM runs.

```
4. Before you answer this one, bring your formal system up to date. It is the thing you are
   building across attempts; the proof is downstream of it. Work in this order, in your
   working directory:

   a. Objects. What things are on the board, and what slots does each thing carry? For every
      object you keep, you should be able to name a pair of positions that stop being
      distinguishable if you drop it.
   b. Rules. Write each rule as a function: given a position and an action, what does it
      change? Each rule needs a domain you can actually execute.
   c. Simplify. Drop slots no rule reads. Merge rules that always decide alike. After this,
      your model's verdict on every frame you have observed must be unchanged.
   d. Formalize. Turn the objects and rules into Lean *definitions* - not theorems. Each rule
      ends up in exactly one of three states: a definition, a consequence of definitions, or
      explicitly marked "not in the model".

   If you already have a formal system from an earlier attempt, **amend it. Do not start a new
   one.** What you learned this attempt is a change to specific objects, rules or definitions;
   make those changes and leave the rest alone. A rewrite throws away the part of your work
   that was already right, and it destroys the record of what this attempt actually taught you.

5. Now prove your answer to question 2 **on that formal system**: that the level can be won,
   or that it cannot.

   - The proof must compile, and `#print axioms` on your main theorem must come back clean -
     no `sorryAx`, no `native_decide`. Report that output verbatim.
   - Say what your theorem actually quantifies over and what it assumes. If the proof needs a
     premise you did not establish, write it as an explicit hypothesis rather than burying it
     in a definition.
   - Name the rules from step 4b that your theorem leans on. If your Lean definitions carry
     something your rules do not - an extra assumption, a tighter domain, a step treated as
     atomic - say so here. A proof that is clean about the wrong system is worth nothing, and
     a model that is *stricter* than the game is the dangerous direction: "impossible in my
     model" does not transfer to the game.
   - Put the file in `lean/` and tell me its filename.
   - If you cannot complete the proof, say exactly where it breaks and what is missing. An
     honest "it does not go through, here is the gap" is worth more than a theorem that proves
     something weaker than it looks.

6. If this attempt taught you nothing that changes your formal system - no rule revised, no
   definition touched - do not invent a change. Say so instead, on its own line, in exactly
   this form, and then give the reason:

       FORMAL-UNCHANGED: <why this attempt added nothing to the model - what you tried, what
       you already knew, and what would have had to happen for the model to change>

   "No change, and here is why" is a real result. A change made only to look busy is not.
```

Verdict lines, AWM runs.

```
End your reply with exactly these two lines, on lines of their own, with nothing after them:

VERDICT: winnable | impossible | impossible_in_model_proved | impossible_in_model | undecided
CONFIDENCE: <an integer from 0 to 100>

The two "in_model" words are for when the level cannot be won under the rules you yourself
measured, and the step from "those rules" to "the real game" is an assumption you are stating
rather than something you established. Say which assumption, in your answer to question 2.
Use ‘impossible_in_model_proved‘ only if you are also handing over a Lean proof of it that
compiles and whose ‘#print axioms‘ is clean - name the file. Otherwise use ‘impossible_in_model‘.
Your proof will be compiled and its axioms checked, so do not claim ‘_proved‘ on a file that
does not go through; ‘impossible_in_model‘ with an honest account of the gap is worth more.

Answer in writing only. Do not send any more game actions.
```

Verdict lines, baseline runs.

```
End your reply with exactly these two lines, on lines of their own, with nothing after them:

VERDICT: winnable | impossible | undecided
CONFIDENCE: <an integer from 0 to 100>

Answer in writing only. Do not send any more game actions.
```

Retry message.

```
You have another attempt on that level.

‘POST /reset‘ restarts the level you were on; the levels you already beat stay beaten, and the
level’s own counters start over. Play it again, as far as you can. When the attempt ends, stop
and report as before.
```

Final interview, questions.

```
That attempt failed.

Answer the following in writing. Give each answer the length it needs; do not pad.

1. What happened on the level you were on, and what do you think made it fail?
   Explain in plain prose, as if to someone who did not watch you play.

2. Do you still see a way to beat this level?
   - If you do: give the line of play, and say what you checked it against.
   - If you no longer do: say why not. State precisely what you are claiming, give the
     argument for it, and make clear what the argument rests on - which parts you established
     and which are assumptions you did not check. Someone else should be able to follow it and
     judge whether it holds. The form of the argument is yours to choose.
   - If you cannot tell either way, say so, and say what would settle it.

3. Which rule did you newly work out across these attempts, and from which observation?

4. Which rule are you still unsure about, and unsure in what way?
```